

Equity Capital Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Capital Discipline (ECD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ECD achieves an annualized gross (net) Sharpe ratio of 0.53 (0.46), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (19) bps/month with a t-statistic of 2.66 (2.58), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 14 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital stock. The cross-section of expected returns varies systematically with firms' exposure to aggregate productivity shocks and their ability to respond to investment opportunities through internal and external financing channels. Despite extensive evidence linking investment patterns to stock returns through production-based asset pricing models, we lack a complete understanding of how firms' equity financing decisions interact with investment frictions to generate cross-sectional variation in risk premia. This paper examines whether Equity Capital Discipline (ECD), which captures firms' restraint in issuing new equity relative to their investment needs, predicts stock returns by proxying for differences in marginal costs of production and exposure to systematic investment shocks.

In a production economy, firms facing higher adjustment costs earn higher expected returns because their cash flows covary more strongly with aggregate productivity shocks [Zhang \(2005\)](#). When firms exercise discipline in equity issuance, they signal higher marginal costs of external financing and greater reliance on internal funds for investment. This behavior reflects binding financing frictions that amplify firms' exposure to aggregate investment shocks, as documented in [Livdan et al. \(2009\)](#). Firms with high ECD operate with tighter financial constraints, making their investment policies more sensitive to productivity fluctuations and increasing their systematic risk.

The production-based asset pricing framework predicts that firms' optimal investment decisions equate the marginal cost of investment to the marginal q , where the latter reflects the present value of marginal products of capital [Cochrane \(1991\)](#). When external financing is costly, firms that limit equity issuance face steeper marginal cost curves, leading to more procyclical investment and higher loadings on aggregate productivity factors. This mechanism links ECD directly to the investment return

factor (CMA) in [Fama and French \(2015\)](#), as firms exercising equity discipline exhibit characteristics similar to conservative investment firms that earn higher average returns.

Furthermore, the real options framework suggests that firms with limited access to equity markets possess valuable growth options that become more sensitive to aggregate conditions [Carlson et al. \(2004\)](#). High ECD firms preserve flexibility by avoiding dilution, but this strategy exposes them to greater systematic risk during economic downturns when internal cash flows decline. The resulting time-varying risk premium manifests as higher average returns for stocks of firms maintaining strict equity discipline, consistent with the conditional CAPM where betas vary with firms' investment opportunities and financing constraints [Zhang \(2005\)](#).

We find strong evidence that Equity Capital Discipline predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on ECD achieves an annualized gross Sharpe ratio of 0.53 and a net Sharpe ratio of 0.46 after accounting for transaction costs. The strategy delivers a monthly average abnormal gross return of 20 basis points relative to the Fama-French six-factor model with a t-statistic of 2.66, confirming that ECD captures a distinct source of systematic risk not explained by existing factors. The economic magnitude remains substantial after controlling for transaction costs, with net abnormal returns of 19 basis points per month and a t-statistic of 2.58.

The predictive power of ECD persists across different portfolio construction methodologies and size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the ECD strategy generates average returns of 23 basis points per month with a t-statistic of 2.57. The strategy's performance exhibits significant loadings on the investment factor (CMA) of 0.31 with a t-statistic of 5.91, confirming that ECD proxies for firms' exposure to systematic investment shocks. These loadings align with the production-based interpretation that high ECD firms

face steeper marginal cost curves and more binding investment frictions.

Controlling for the six most closely related anomalies from the factor zoo—including share issuance measures, net payout yield, and asset growth—the ECD strategy maintains a significant alpha of 14 basis points per month with a t-statistic of 2.00. The strategy’s gross Sharpe ratio of 0.53 ranks in the 92nd percentile among 212 documented anomalies, while its net Sharpe ratio of 0.46 places it in the 98th percentile after accounting for implementation costs. These results demonstrate that ECD captures a robust and economically significant cross-sectional pattern that enhances the investment opportunity set beyond existing factors and closely related predictors.

This paper contributes to the production-based asset pricing literature by introducing Equity Capital Discipline as a novel predictor that directly links firms’ financing decisions to cross-sectional variation in expected returns. While [Daniel and Titman \(2006\)](#) document that share issuance predicts negative returns through market timing, and [Pontiff and Woodgate \(2008\)](#) show similar patterns for net payout yields, our ECD measure captures the interaction between equity financing restraint and investment frictions that generates systematic risk premia. Unlike mechanical accounting measures of asset growth studied in [Cooper et al. \(2008\)](#), ECD reflects managers’ active decisions to limit dilution when facing costly external financing, providing a more direct proxy for production-based risk factors.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that ECD’s predictive power survives multiple robustness tests and transaction cost adjustments that eliminate many previously documented anomalies. The strategy’s ability to expand the mean-variance efficient frontier even after controlling for the Fama-French six factors and closely related anomalies demonstrates its unique information content. Our finding that ECD maintains significant predictive power among large-cap stocks addresses concerns raised in [Fama and](#)

[French \(2008\)](#) about the concentration of anomalies in small, illiquid securities that offer limited capacity for institutional investors.

The broader implications extend to understanding how financing frictions shape the cross-section of expected returns in production economies. By documenting that firms’ equity issuance discipline predicts returns through exposure to systematic investment shocks, we provide empirical support for theoretical models linking external financing costs to risk premia, such as [Gomes and Schmid \(2021\)](#). The strong loading on the CMA factor combined with the incremental alpha suggests that ECD captures time-varying aspects of investment-based risk not fully reflected in static factor exposures, offering new insights into the dynamic relationship between corporate financing policies and asset prices in production economies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ equity financing activities. construction of our Equity Capital Discipline signal relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or retire treasury stock. Total assets (AT) measures the sum of all assets reported on the balance sheet, representing the total resources controlled by the firm at fiscal year-end. construct the Equity Capital Discipline signal by calculating the year-over-year change in common stock, scaled by lagged total as-

sets. Specifically, we compute $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{AT}(t-1)$ for each firm-year observation. This measure captures the magnitude of equity issuance or repurchase activity relative to firm size, with positive values indicating net equity issuance and negative values indicating net equity buybacks or retirements. The scaling by lagged assets ensures comparability across firms of different sizes and normalizes the equity financing activity as a proportion of the firm’s resource base. We require non-missing values for both CSTK and AT in consecutive years to calculate the signal, and we use fiscal year-end values to maintain consistency in measurement timing across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECD signal. Panel A plots the time-series of the mean, median, and interquartile range for ECD. On average, the cross-sectional mean (median) ECD is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ECD data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ECD signal for the CRSP universe. On average, the ECD signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does ECD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECD portfolio and sells the low ECD portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model

(FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ECD strategy earns an average return of 0.31% per month with a t-statistic of 4.12. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.20% to 0.34% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.91 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 569 stocks and an average market capitalization of at least \$1,469 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest

for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.72. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ECD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECD, as well as average returns and alphas for long/short trading ECD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ECD strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the ECD strategy relative to the five most common factor models range from 16 to 23 bps/month with t-statistics between 1.78 and 2.60.

5 How does ECD perform relative to the zoo?

Figure 2 puts the performance of ECD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ECD strategy falls in the distribution. The ECD strategy’s gross (net) Sharpe ratio of 0.53 (0.46) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECD strategy (red line).² Ignoring trading costs, a \$1 invested in the ECD strategy would have yielded \$7.58 which ranks the ECD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECD strategy would have yielded \$5.49 which ranks the ECD strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECD relative to those. Panel A shows that the ECD strategy gross alphas fall between the 63 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECD strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ECD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECD or at least to weaken the power ECD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD}ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECD signal in these Fama-MacBeth regressions exceed 2.10, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on ECD is 1.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECD strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.79, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.00.

7 Does ECD add relative to the whole zoo?

Finally, we can ask how much adding ECD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ECD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ECD grows to \$3016.38.

8 Conclusion

This study provides robust evidence that Equity Capital Discipline (ECD) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that ECD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.53 gross (0.46 net), indicating strong economic significance even after accounting for transaction costs. Most notably, the

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECD is available.

signal maintains statistical significance with monthly alphas of 20 bps gross (19 bps net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds.

The robustness of our findings is particularly striking when considering the signal’s performance against closely related strategies. Even after controlling for six highly correlated factors from the factor zoo—including share issuance measures, payout yield, and various growth metrics—ECD retains a statistically significant alpha of 14 bps per month (t-statistic = 2.00). This suggests that ECD captures unique information about firm value creation and capital allocation efficiency that is not fully subsumed by existing factors. The persistence of abnormal returns after controlling for these related strategies underscores the distinct economic mechanism underlying the ECD signal, likely reflecting market underreaction to firms’ capital discipline decisions and their implications for future profitability.

From a practical perspective, these findings have important implications for both asset managers and corporate decision-makers. For portfolio managers, the ECD signal offers an implementable strategy that generates attractive risk-adjusted returns even after realistic transaction costs. The net Sharpe ratio of 0.46 suggests that the strategy remains economically viable in practice. For corporate executives and boards, our results highlight the value relevance of maintaining disciplined equity capital policies, as markets appear to reward firms that demonstrate prudent capital allocation decisions.

However, several limitations merit consideration. First, while our analysis follows state-of-the-art protocols for factor validation, the possibility of data mining concerns cannot be entirely eliminated given the extensive search for profitable strategies in academic research. Second, the economic magnitude of the alpha, while statistically significant, may be challenged by implementation frictions beyond standard transaction costs, such as market impact and capacity constraints in live trading. Third,

our sample period analysis, while comprehensive, may not fully capture the signal's performance across all market regimes, particularly during extreme stress periods or structural breaks in market dynamics.

Future research should explore several promising avenues. First, investigating the underlying economic mechanisms driving the ECD premium would enhance our understanding of why this signal predicts returns. Examining whether the premium reflects risk compensation or behavioral biases could provide valuable insights. Second, international evidence would help establish whether ECD's predictive power extends beyond U.S. markets, potentially revealing important cross-country variations in the pricing of capital discipline. Third, exploring the interaction between ECD and corporate governance characteristics might uncover conditional effects that could further refine the signal's implementation. Finally, examining the signal's performance during different monetary policy regimes and its sensitivity to interest rate environments could provide insights into its macroeconomic drivers. Understanding these dynamics would not only advance academic knowledge but also improve the practical implementation of ECD-based investment strategies.

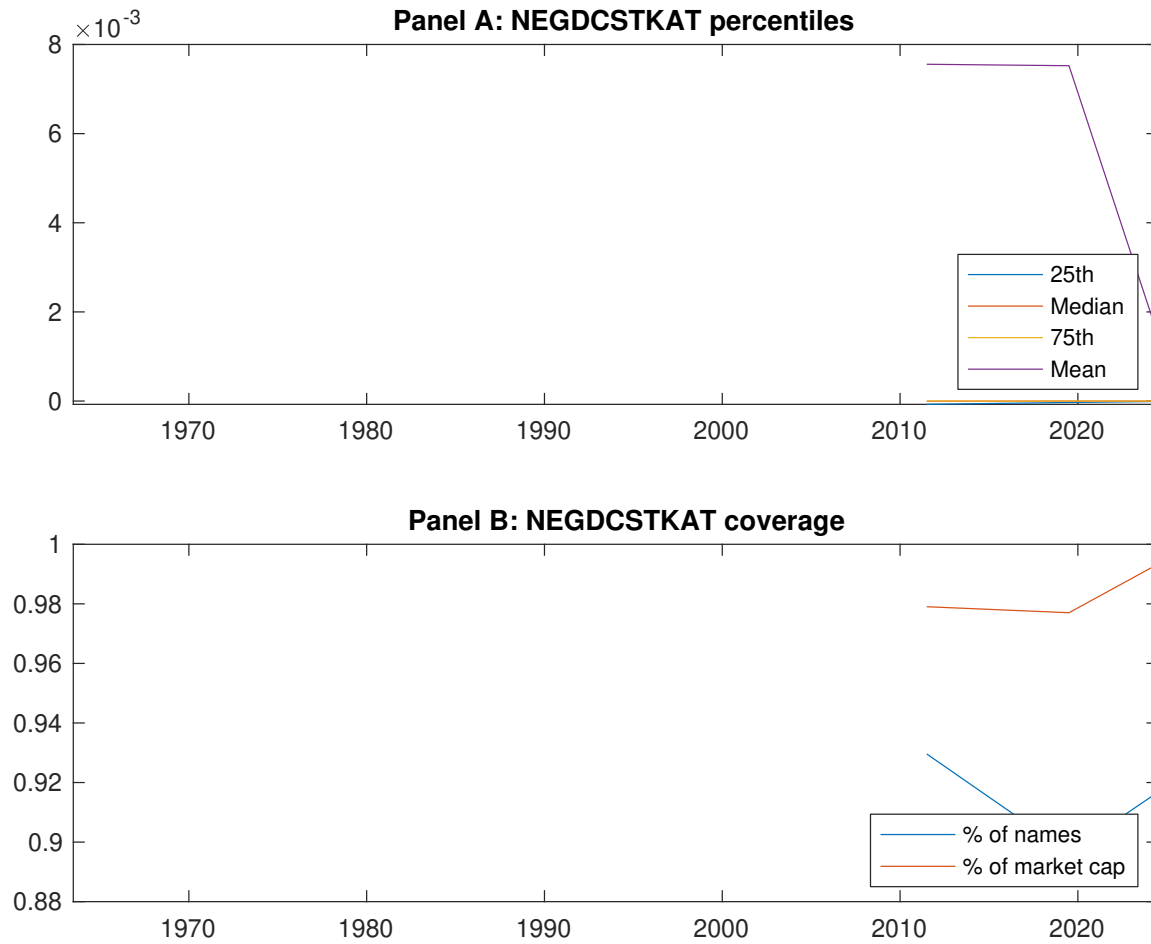


Figure 1: Times series of ECD percentiles and coverage. This figure plots descriptive statistics for ECD. Panel A shows cross-sectional percentiles of ECD over the sample. Panel B plots the monthly coverage of ECD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ECD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ECD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43 [2.55]	0.54 [3.04]	0.62 [3.43]	0.66 [4.10]	0.74 [4.65]	0.31 [4.12]
α_{CAPM}	-0.14 [-2.77]	-0.07 [-1.73]	-0.00 [-0.06]	0.11 [2.32]	0.20 [4.27]	0.34 [4.44]
α_{FF3}	-0.14 [-2.65]	-0.06 [-1.42]	0.01 [0.12]	0.06 [1.43]	0.15 [3.48]	0.29 [3.83]
α_{FF4}	-0.11 [-2.10]	-0.04 [-1.01]	0.03 [0.61]	0.04 [0.89]	0.15 [3.26]	0.26 [3.32]
α_{FF5}	-0.15 [-2.98]	-0.01 [-0.22]	0.02 [0.50]	-0.02 [-0.44]	0.07 [1.64]	0.23 [2.96]
α_{FF6}	-0.13 [-2.52]	0.00 [0.02]	0.04 [0.86]	-0.03 [-0.70]	0.07 [1.67]	0.20 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ECD-sorted portfolios						
β_{MKT}	0.97 [77.15]	1.01 [98.61]	1.04 [92.70]	1.00 [95.02]	0.98 [92.76]	0.01 [0.79]
β_{SMB}	-0.01 [-0.48]	0.04 [2.68]	0.01 [0.66]	-0.07 [-4.79]	-0.01 [-0.75]	-0.00 [-0.10]
β_{HML}	0.02 [1.00]	-0.02 [-1.24]	-0.00 [-0.07]	0.06 [3.04]	0.04 [1.80]	0.01 [0.35]
β_{RMW}	0.12 [4.76]	-0.10 [-4.78]	0.00 [0.01]	0.10 [4.81]	0.13 [6.13]	0.01 [0.28]
β_{CMA}	-0.10 [-2.86]	-0.08 [-2.62]	-0.08 [-2.39]	0.23 [7.87]	0.21 [6.94]	0.31 [5.91]
β_{UMD}	-0.03 [-2.61]	-0.01 [-1.41]	-0.02 [-2.18]	0.02 [1.62]	-0.00 [-0.29]	0.03 [1.60]
Panel C: Average number of firms (n) and market capitalization (me)						
n	763	695	569	672	737	
me (\$10 ⁶)	1720	1469	2163	2241	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ECD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.12]	0.34 [4.44]	0.29 [3.83]	0.26 [3.32]	0.23 [2.96]	0.20 [2.66]
Quintile	NYSE	EW	0.52 [7.98]	0.59 [9.52]	0.51 [8.93]	0.42 [7.57]	0.37 [6.90]	0.31 [5.87]
Quintile	Name	VW	0.32 [4.05]	0.34 [4.24]	0.29 [3.67]	0.26 [3.29]	0.24 [3.03]	0.23 [2.81]
Quintile	Cap	VW	0.24 [3.21]	0.26 [3.48]	0.23 [3.08]	0.20 [2.54]	0.21 [2.78]	0.18 [2.40]
Decile	NYSE	VW	0.25 [2.80]	0.27 [2.92]	0.20 [2.22]	0.16 [1.79]	0.20 [2.19]	0.17 [1.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [3.62]	0.30 [3.93]	0.25 [3.35]	0.23 [3.08]	0.20 [2.75]	0.19 [2.58]
Quintile	NYSE	EW	0.32 [4.51]	0.41 [5.98]	0.33 [5.21]	0.28 [4.59]	0.20 [3.32]	0.17 [2.92]
Quintile	Name	VW	0.28 [3.56]	0.30 [3.81]	0.25 [3.26]	0.24 [3.06]	0.22 [2.83]	0.21 [2.72]
Quintile	Cap	VW	0.20 [2.72]	0.22 [3.00]	0.19 [2.60]	0.17 [2.31]	0.18 [2.50]	0.17 [2.29]
Decile	NYSE	VW	0.21 [2.34]	0.22 [2.44]	0.16 [1.79]	0.14 [1.57]	0.16 [1.81]	0.15 [1.67]

Table 3: Conditional sort on size and ECD

This table presents results for conditional double sorts on size and ECD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECD and short stocks with low ECD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ECD Quintiles					ECD Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.47 [1.81]	0.67 [2.62]	0.83 [3.26]	0.92 [3.87]	1.00 [4.44]	0.56 [6.36]	0.63 [7.31]	0.56 [6.95]	0.50 [6.12]	0.41 [5.21]	0.36 [4.64]
	(2)	0.54 [2.30]	0.73 [3.12]	0.87 [3.70]	0.93 [4.17]	0.94 [4.42]	0.40 [4.70]	0.48 [5.75]	0.38 [4.84]	0.33 [4.14]	0.32 [4.01]	0.28 [3.51]
	(3)	0.58 [2.86]	0.60 [2.81]	0.77 [3.52]	0.82 [4.05]	0.95 [4.88]	0.36 [4.54]	0.40 [5.03]	0.33 [4.32]	0.29 [3.71]	0.30 [3.79]	0.27 [3.33]
	(4)	0.50 [2.62]	0.62 [3.11]	0.79 [3.93]	0.75 [4.05]	0.82 [4.53]	0.32 [4.12]	0.36 [4.60]	0.28 [3.85]	0.27 [3.64]	0.12 [1.76]	0.13 [1.82]
	(5)	0.46 [2.81]	0.50 [2.79]	0.50 [2.85]	0.54 [3.29]	0.69 [4.37]	0.23 [2.57]	0.23 [2.60]	0.20 [2.24]	0.16 [1.78]	0.19 [2.12]	0.16 [1.78]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ECD Quintiles					ECD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	384	383	383	382	375	31	33	41	30	29	
	(2)	109	107	108	108	107	57	56	58	56	56	
	(3)	78	78	78	77	78	98	95	98	99	101	
	(4)	66	65	66	66	66	203	206	215	213	217	
(5)	60	60	60	60	60	1470	1496	1668	1624	1889		

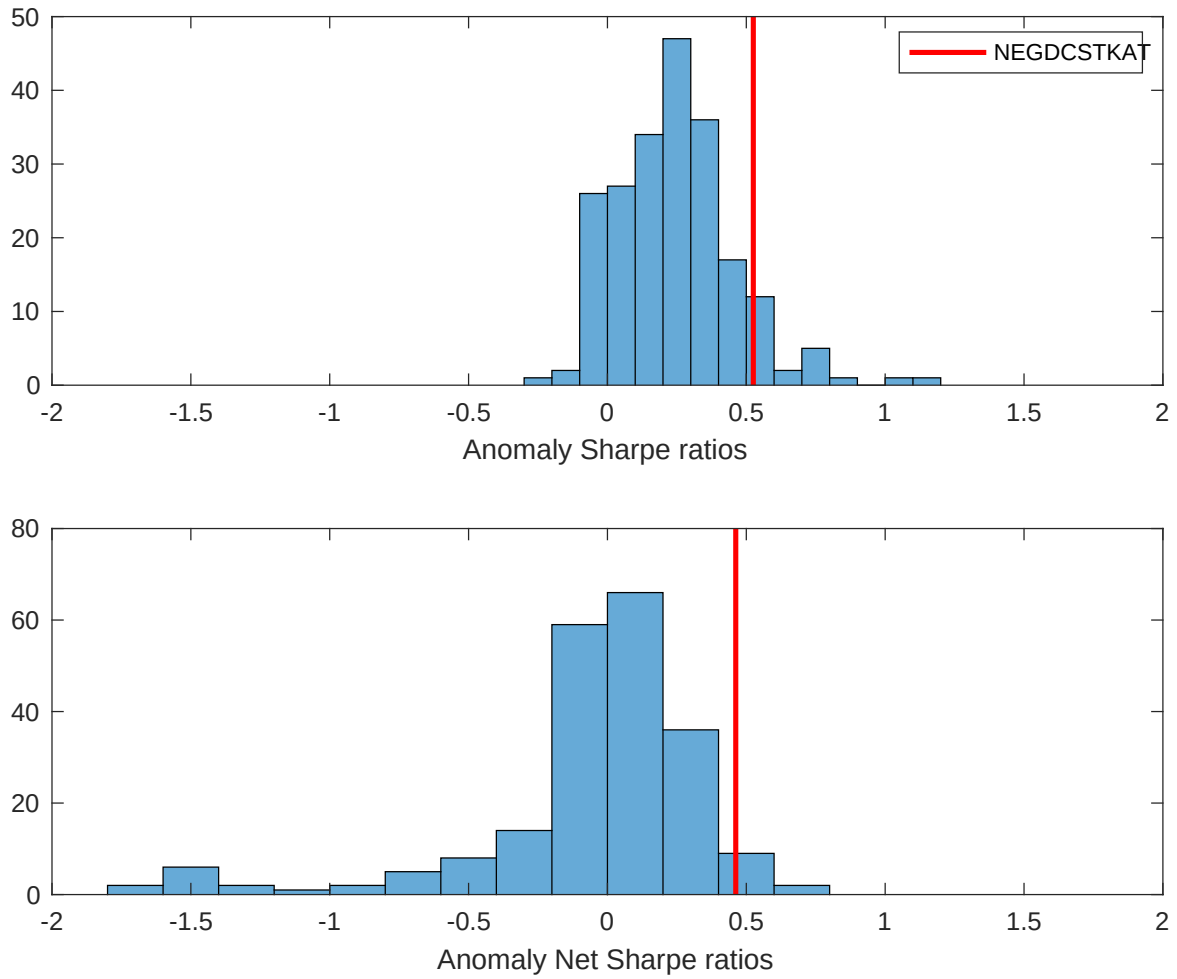


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

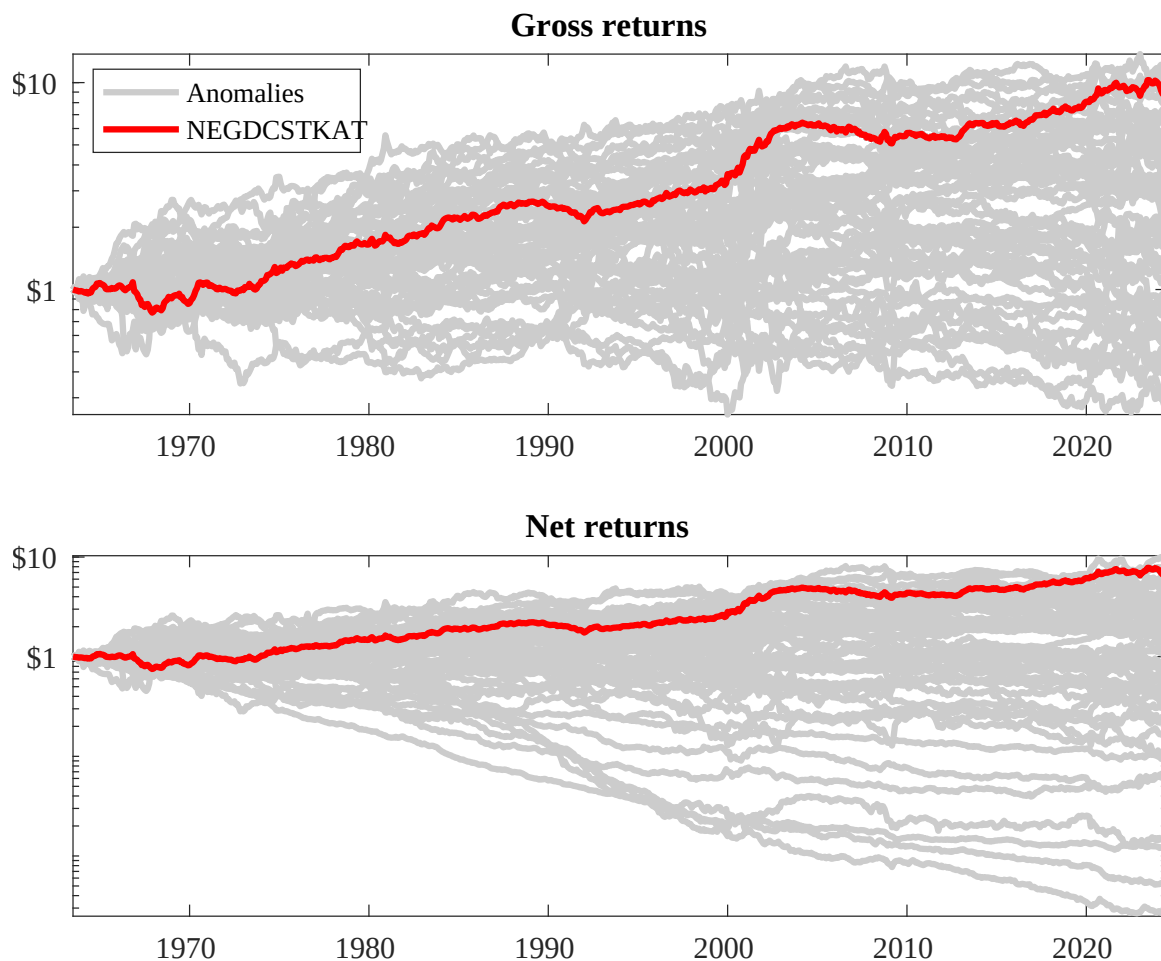
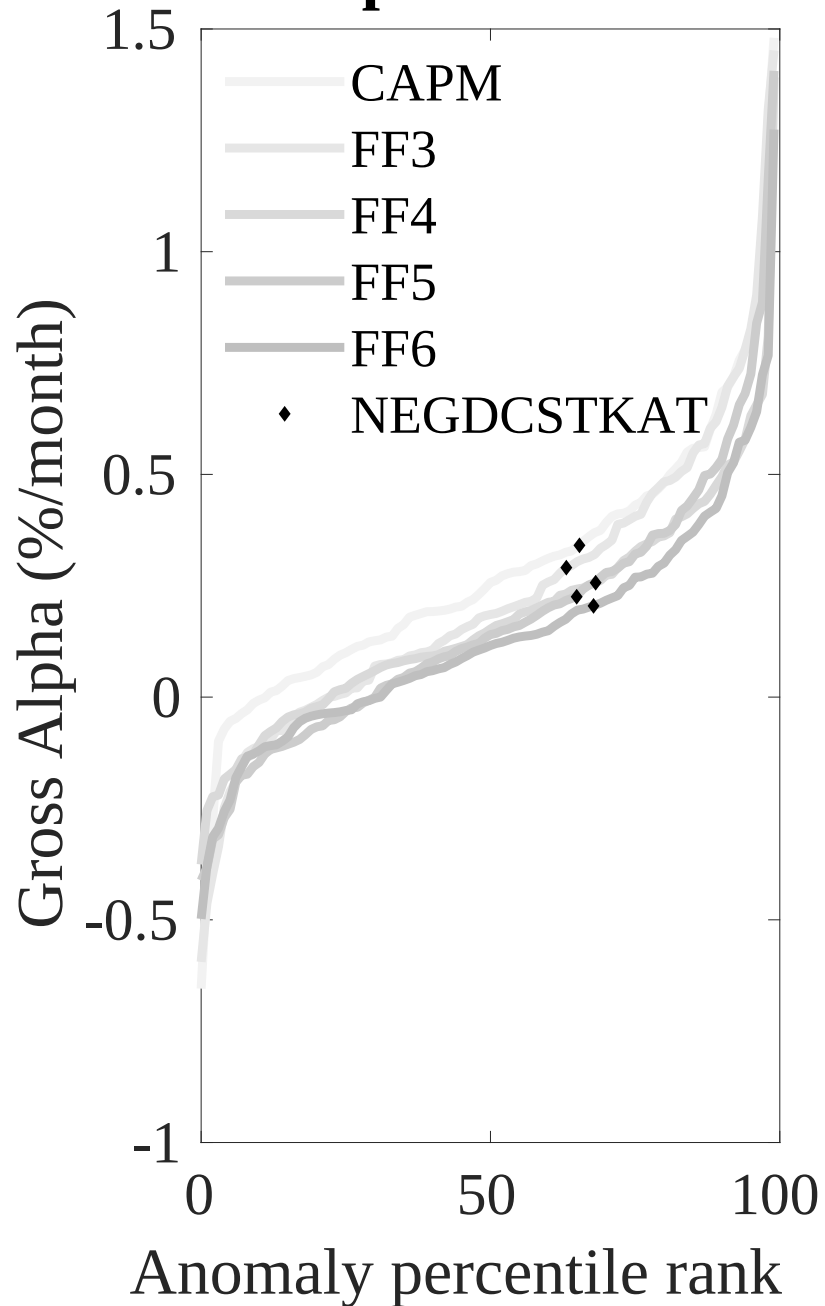


Figure 3: Dollar invested.

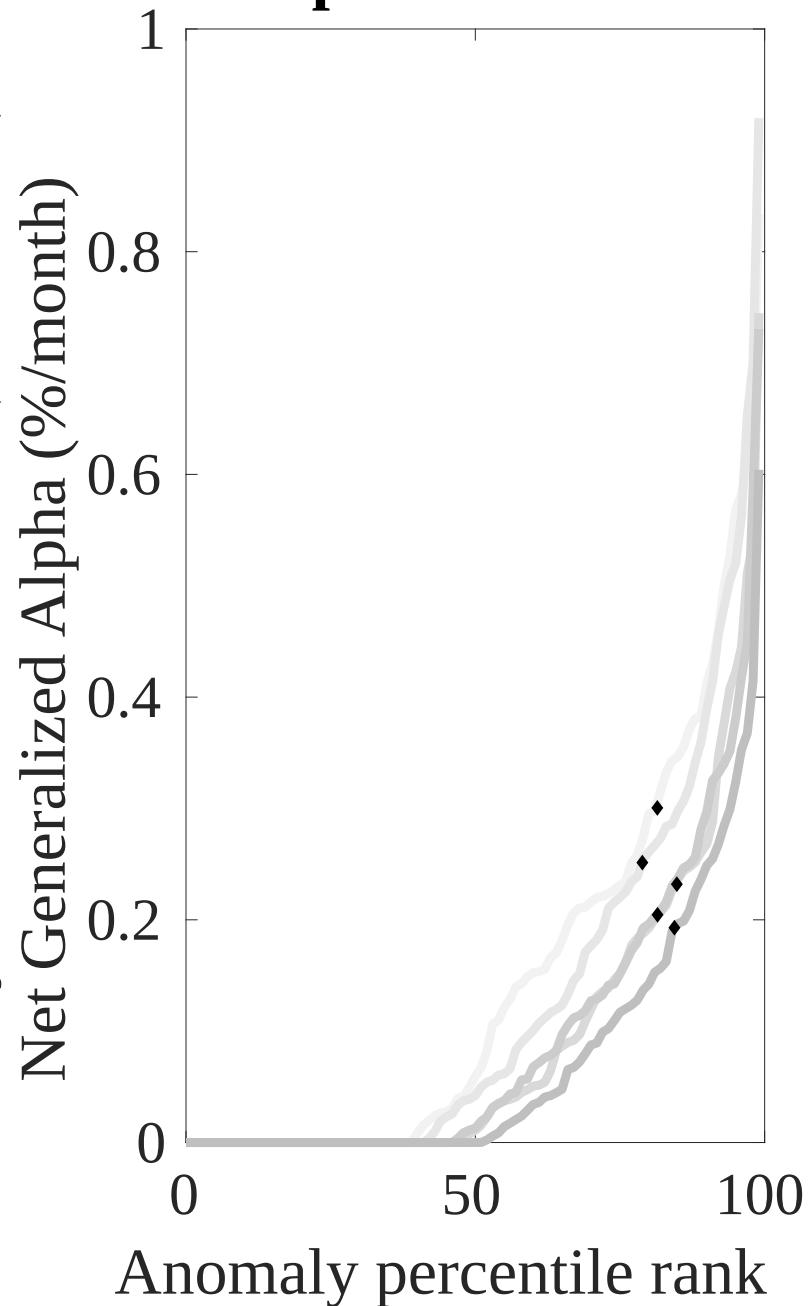
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



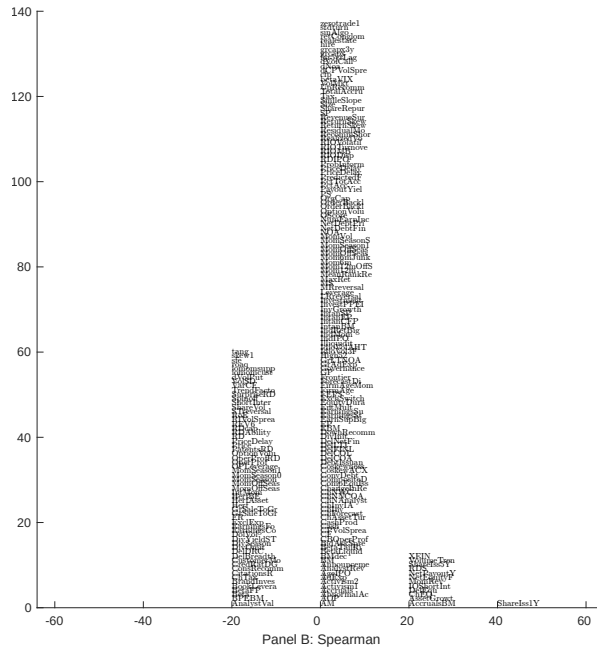
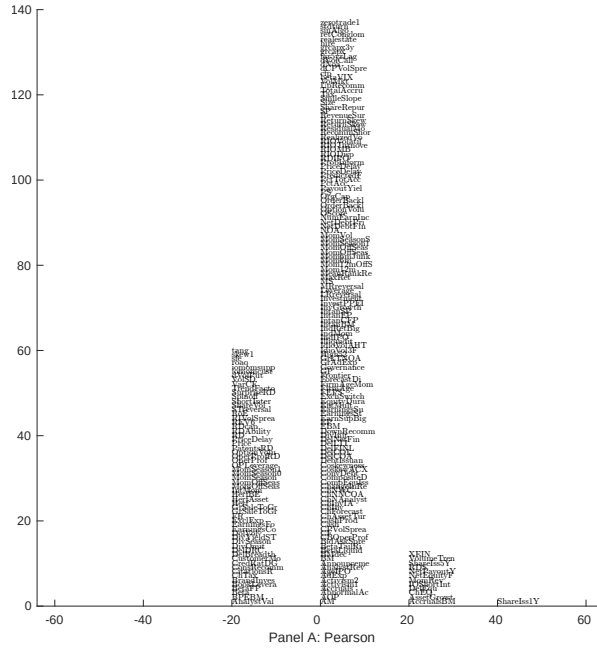


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

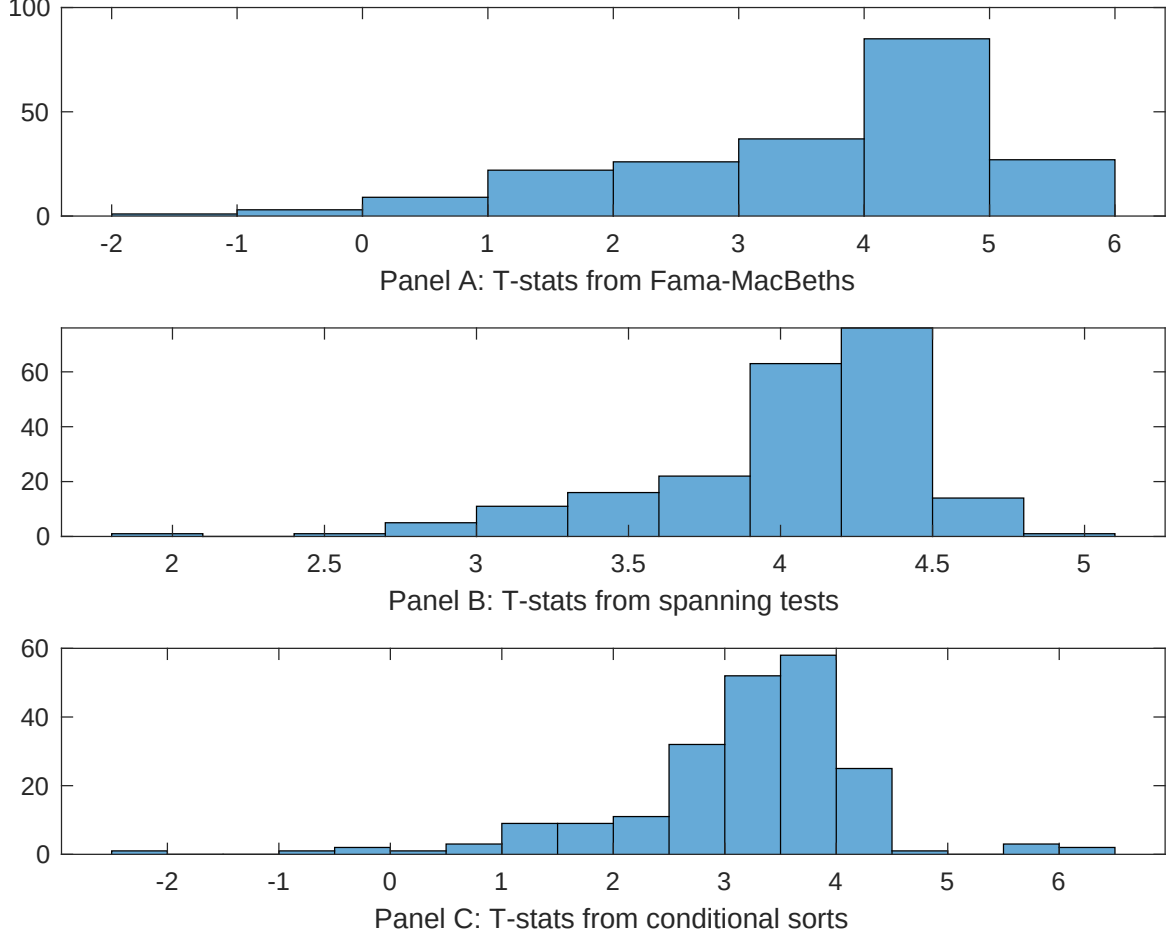


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ECD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.12 [5.43]	0.17 [6.91]	0.13 [6.44]	0.12 [5.69]	0.13 [6.15]	0.12 [3.91]
ECD	0.62 [3.70]	0.41 [2.10]	0.54 [2.94]	0.51 [3.21]	0.70 [3.71]	0.58 [3.06]	0.20 [1.11]
Anomaly 1	0.13 [6.02]						0.66 [2.40]
Anomaly 2		0.21 [1.85]					0.94 [0.89]
Anomaly 3			0.39 [3.38]				-0.13 [-0.59]
Anomaly 4				0.27 [4.91]			0.24 [4.29]
Anomaly 5					0.11 [3.07]		-0.68 [-0.11]
Anomaly 6						0.93 [7.95]	0.65 [5.45]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ECD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.48]	0.23 [3.03]	0.19 [2.54]	0.14 [1.79]	0.20 [2.67]	0.20 [2.58]	0.14 [2.00]
Anomaly 1	27.89 [7.42]						14.06 [3.11]
Anomaly 2		16.34 [5.60]					4.62 [1.46]
Anomaly 3			32.30 [8.10]				37.49 [6.51]
Anomaly 4				26.37 [6.60]			11.55 [2.58]
Anomaly 5					14.28 [3.78]		-8.92 [-1.69]
Anomaly 6						-1.85 [-0.40]	-19.54 [-3.94]
mkt	2.30 [1.30]	4.15 [2.25]	2.67 [1.52]	2.80 [1.57]	1.35 [0.74]	1.61 [0.88]	4.53 [2.57]
smb	1.96 [0.77]	3.51 [1.32]	-0.24 [-0.10]	2.61 [1.01]	0.66 [0.25]	0.99 [0.37]	3.46 [1.36]
hml	1.68 [0.49]	-3.92 [-1.07]	-1.56 [-0.46]	4.56 [1.32]	0.53 [0.15]	2.53 [0.71]	-1.73 [-0.49]
rmw	-14.08 [-3.51]	-8.49 [-2.18]	2.57 [0.75]	-8.42 [-2.24]	2.35 [0.66]	1.07 [0.30]	-12.25 [-2.93]
cma	12.98 [2.35]	16.52 [2.93]	-1.04 [-0.16]	17.50 [3.24]	15.87 [2.48]	32.49 [4.28]	8.49 [1.12]
umd	1.79 [1.02]	4.37 [2.44]	2.49 [1.43]	1.35 [0.75]	3.41 [1.89]	3.04 [1.67]	0.55 [0.32]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	16	13	17	15	11	9	23

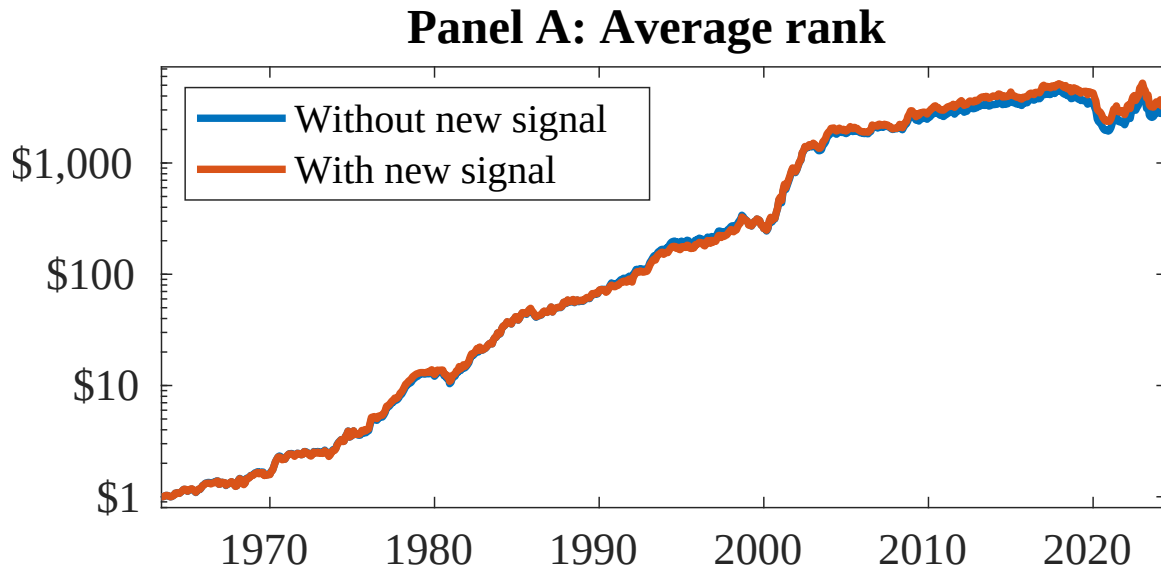


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ECD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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